

Data Platform Capabilities Technical Architecture & Analytics Framework

Generated Report

March 09, 2026

Content

Executive Summary

- **Database:** PostgreSQL with 3,156,332 rows across 4 fact tables
- **Coverage:** 199 equities (12 sectors) | 48 crypto assets | 106 cities (35 states) | 49 macro indicators
- **Pipeline:** Automated Python scripts pulling from Yahoo Finance, FRED, Open-Meteo
- **Analytics:** 20+ KPIs computed per report including Sharpe, VaR, beta, correlation matrices

Data Architecture

Four fact tables in PostgreSQL provide the analytical foundation. All data is pulled from public, freely-available APIs with full audit trail.

fact_equity_daily

Component	Detail
Source	Yahoo Finance Chart API (free, no key)
Rows	249,944
Coverage	199 tickers across 12 GICS sectors
Date Range	2021-03-08 to 2026-03-06
Columns	ticker, company_name, sector, trade_date, open, high, low, close_price, volume
Indexes	ticker, trade_date, sector

fact_crypto_daily

Component	Detail
Source	Yahoo Finance (crypto tickers: BTC-USD, ETH-USD, etc.)
Rows	33,980
Coverage	48 crypto assets
Date Range	2024-03-08 to 2026-03-08
Columns	coin_id, coin_name, symbol, trade_date, price_usd, total_volume_usd
Indexes	coin_id, trade_date, symbol

fact_weather_hourly

Component	Detail
Source	Open-Meteo Archive API (free, no key)
Rows	2,788,224
Coverage	106 US cities across 35 states

Component	Detail
Date Range	2023-01-01 to 2025-12-31
Columns	city, state, lat, lon, observation_time, temperature_c, humidity, precipitation_mm, wind_speed_kmh, surface_pressure_hpa
Indexes	city, observation_time, state

fact_macro_monthly

Component	Detail
Source	FRED (Federal Reserve Economic Data) — CSV download
Rows	84,184
Coverage	49 economic series (GDP, CPI, rates, employment, housing, commodities, forex)
Date Range	2005-01-01 to 2026-03-06
Columns	series_id, series_name, category, observation_date, value, units, frequency
Indexes	series_id, observation_date, category

KPI Computation Framework

All KPIs are computed from raw data using standard financial formulas. No simulated or estimated values.

Return Metrics

KPI	Formula / Value	Interpretation
Period Return	$(P_{end} - P_{start}) / P_{start}$	Simple price return over any period (week, month, quarter, YTD)
Weighted Return	$\text{Sum}(\text{weight}_i * \text{return}_i)$	Volume-weighted (crypto) or equal-weighted portfolio return
Week-over-Week	$WK_return - PREV_WK_return$	Momentum signal — acceleration or deceleration of returns

Risk Metrics

KPI	Formula / Value	Interpretation
Sharpe Ratio	$(\text{Ann_Return} - R_f) / \text{Ann_Vol}$	Risk-adjusted return. $R_f = 4.5\%$ (T-bill proxy). Annualized from daily.
Sortino Ratio	$(\text{Ann_Return} - R_f) / \text{Downside_Dev}$	Penalizes only downside volatility. More relevant for crypto.
VaR (95%)	$\mu - 1.645 * \sigma$	Parametric variance-covariance. Maximum expected daily loss at 95% confidence.
Max Drawdown	$\min((P - \text{Peak}) / \text{Peak})$	Largest peak-to-trough decline within the period.
Annualized Volatility	$\sqrt{\text{sum}(r^2)/n} * \sqrt{365}$	Root-mean-square of daily returns, annualized.

Portfolio Metrics

KPI	Formula / Value	Interpretation
Beta	$\text{Cov}(R_{\text{asset}}, R_{\text{market}}) / \text{Var}(R_{\text{market}})$	Systematic risk vs benchmark (SPY). $\text{Beta} > 1$ = higher market sensitivity.
R-Squared	$\text{Correlation}(R_{\text{asset}}, R_{\text{market}})^2$	Proportion of variance explained by market. High R^2 = index-like behavior.
Alpha (Jensen)	$\text{Return} - [R_f + \text{Beta} * (\text{Market_Return} - R_f)]$	Excess return beyond what beta explains. Positive = outperformance.
HHI	$\text{sum}(\text{weight}_i^2) * 10000$	Herfindahl-Hirschman Index. >2500 = concentrated, <1500 = diversified.
Top 3 Concentration	$\text{sum}(\text{top 3 weights})$	Portfolio weight in three largest positions.
Correlation Matrix	$\text{Cov}(A,B) / (\text{std}_A * \text{std}_B)$	Pairwise return correlations. Computed from daily returns over analysis period.

Cross-Source Analytics

The platform combines data from independent sources to create richer analytical products.

Demonstrated Cross-Source Combinations

Component	Detail
Energy + Weather	Oil & gas equity performance correlated with temperature and precipitation in producing regions (Houston, Midland, Denver)
Energy + Macro	WTI crude and Henry Hub natural gas prices (FRED) alongside energy equity returns (Yahoo Finance)
Sports + Weather	NFL game-day conditions (Open-Meteo) at venue cities combined with sports media equity performance
Finance + Macro	Equity portfolio analysis enriched with Fed Funds rate, 10Y yield, CPI, and unemployment context

Report Generation Pipeline

- Data Hydration:** Python scripts pull from public APIs into PostgreSQL fact tables (one-time + scheduled refresh)
- Query Layer:** SQL queries with parameterized date ranges extract relevant data for each cadence
- KPI Engine:** Python computes all metrics from raw data — no pre-aggregated tables
- HTML Generation:** Shared template module produces consistent, branded report layout
- PDF Rendering:** WeasyPrint converts HTML to print-ready PDF with headers, footers, page breaks
- Quality Assurance:** Row counts in report footer must match database. KPIs spot-checked against sources.

Vertical Coverage

Risk Factor	Level	Commentary
Finance	ACTIVE	199 equities, 12 sectors, Sharpe/beta/alpha/VaR
Crypto	ACTIVE	48 digital assets, volume-weighted, Sortino/HHI/correlation
Energy	ACTIVE	16 energy equities + 2 FRED commodity series + 10 weather cities
Sports	ACTIVE	6 sports/media equities + 10 NFL venue weather stations

This report uses **real data from Yahoo Finance, FRED, and Open-Meteo, stored in PostgreSQL**. "Data Platform Capabilities" is a fictional firm name used for demonstration — all underlying market data is real and independently verifiable. This report does not constitute financial advice, investment recommendation, or solicitation. Past performance does not guarantee future results.

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Data Source: PostgreSQL (3,156,332+ rows) | Processing: Clean Metrics Studio (CMS)

Clean Metrics Studio